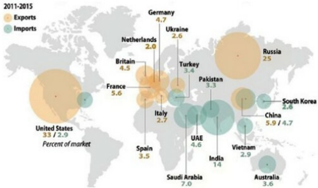

Fig. 1: World's largest exporters and importers of weapons

country's GDP is spent on defence. With the increased threats at our borders, the spend on defence is slated to touch 7-8 per cent of the GDP in the next five years. That's the reason why Make in India is critical for the defence sector in times to come. In the process, traditional manufacturing methods would need to give way to lean and green options, even while improving the bottomline for businesses. Capital investment estimates for India's defence sector over the next five years hint at a figure of US$ 250 billion, with 53 per cent invested in the Army, 23 per cent in the Air Force, 16 per cent in the Navy, 6 per cent in defence R&D, 1 per cent on ordinance factory development and the remaining 1 per cent on other miscellaneous activities.

India today gets nearly 70 per cent of its defence requirements from abroad, but that's slated to change with the new directives coming into force. And with the FDI level opening up to 49 per cent, the sector is surely going to see changes, with projections that Indian manufacturing for the defence sector will grow from 15 per cent of the total market to 25 per cent within five years.

So let's look at the major products that are being made in India for its armed forces today, by adopting indigenous and green processes that have a focus on reusability.

Sonar dome: India has joined a select group of countries that have the capabilities to manufacture the sonar dome. This dome is attached to the bottom of ships, and scans the seas for submarine threats. The manufacturing requirements are technically demanding.

The sonar dome

The R&D for this project was done at Pune, by the DRDO lab under the aegis of Research and Development Establishment Engineers (RDEE). The dome was manufactured by Kineco Ltd, a composites manufacturing facility in Goa. The other critical component in the design, Vacuum Assisted Resin Transfer Moulding (VARTM), was also successfully manufactured in India. The developments shown in this regard stand to benefit the land based and aerospace applications in the future.

Varunastra torpedo: This underwater missile has been developed by research scientists at the Naval Science and Technological Labs (NTSL) in partnership with the Hyderabad based

Manohar Parrikar, former defence minister, takes a look at the indigenously designed Varunastra torpedo

PSU, Bharat Dynamics, with a defined road map from creation to commissioning. And 2016 was the year when this initiative turned into reality.

At the time of its launch and commissioning to the services,